QP CODE: 24900204

SERIES: A

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS) REGULAR
EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - **MG1MDCPHY100 - PHYSICS AROUND YOU**

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A

Multiple Choice Questions

Answer all questions

Each carries **1 mark**

1. What is the SI unit of temperature? [K] [1]

   Ans: **c) Kelvin**

2. 1 Litre = _____ millilitres [K] [1]

   Ans: **c) 1000**

3. What is velocity? [U] [1]

   Ans: **a) Speed with direction**

4. In free fall, what remains constant for a falling object? [U] [1]

   Ans: **c) Acceleration due to gravity**

5. What is the unit of Efficiency? [K] [1]

   Ans: **c) No Unit**

6. When the source of a sound is moving towards a stationary observer, [U] [1]
   the frequency of the sound:

   Ans: **a) Increases**

7. Wavelength range of sound waves audible to humans is: [U] [1]

   Ans: **a) 20 to 20000 Hz**

8. Electric power is the product of ⋯..... [K] [2]

   Ans: **c) Current and Voltage**

9. A circuit that has only one path for current to flow is called: [U] [2]

   Ans: **a) Series circuit. Give credit to <u>any answer</u> as the Question is out of syllabus**

   <u>Also, use the correct CO as CO2 (given as CO1 in the question paper)</u>

10. Which material is commonly used in making optical fibers? [K] [3]

    Ans: **a) Plastic** OR **b) Glass**

11. Which phenomenon explains the whiteness of clouds? [K] [3]

Ans: **b) Mie scattering. Give credit to <u>any answer</u> as the Question is out of syllabus**

12. Why does blue light scatter more than red light? [U] [3]

Ans: **b) Shorter wavelength**

13. Which phenomenon makes stars appear to flicker? [U] [3]

Ans: **c) Refraction**

14. Which application uses total internal reflection? [A] [3]

Ans: **a) Endoscopes. Give credit to <u>any answer</u> as the Question is out of syllabus**

15. Laser light is produced due to [U] [4]

Ans: **c) stimulated emission of radiation**

(15 × 1 = 15)

Part B

Multiple Choice Questions
Answer all questions
Each question carries **2 marks**

16. Which law of motion states that force is equal to mass times acceleration? [K] [1]

Ans: **b) Newton's Second Law**

17. A 60 kg person climbs 5 m up a staircase. What is the work done? [U] [1]

Ans: **a) 2940 J**

18. A crane lifts a load of 200 kg at 5 m/s by applying a force of 1000 N [U] [1]

Ans: **a) 5000 W**

19. What is the primary characteristic that distinguishes longitudinal waves from transverse waves? [U] [1]

Ans: **d) Direction of particle displacement**

20. In astronomy, the Doppler effect is used to measure: [U] [1]

Ans: **b) The velocity of stars and galaxies**

21. What is the frequency of AC supply in India? [K] [2]

Ans: **b) 50 Hz**

22. What is the potential difference across a resistor of 2Ω with 5A current? [U] [2]

Ans: **a) 10V**

23. Waves in decreasing order of their wavelength are [K] [3]

Ans: **c) Radio waves, infrared rays, visible light, X - Rays**

24. What is the critical angle in optics? [K] [3]

Ans: **c) The angle beyond which total internal reflection occurs**

25. What is the primary function of the lasing medium in a laser system? [U] [4]

Ans: **b) Amplify light**

(10 × 2 = 20)

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